DINETH WICKRAMASINGHE

AI Application Engineer Intern

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SUMMARY

I am experienced in building agentic LLM pipelines, RAG systems, and production-ready Python backends using FastAPI and LangChain. Proficient in multi-agent orchestration, vector and graph databases (Qdrant, Neo4j), and Docker containerization to transform complex AI logic into reliable software applications.

PROFESSIONAL EXPERIENCE

Department Of Examinations - Sri Lanka

AI Application Engineer Intern (Volunteer) 08/2026 – Present | Pelawatta, Sri Lanka

- Refine the automated validation workflow by engineering strict context validation logic, ensuring generated assessment items maintain factual alignment with official syllabus materials.
- Optimize the hybrid retrieval process across vector and knowledge graph layers, reducing query latency and maximizing context precision during test generation.

AI Application Engineer Intern 02/2026 – 08/2026 | Pelawatta, Sri Lanka

- Architected a multi-agent LLM pipeline on Dify featuring a 4-workflow system (Coordinator, Context, Generation, Validation) with Redis-backed state management to prevent payload corruption during tool calls.
- Engineered a self-correcting retry loop that automatically regenerates failed assessment questions based on structured feedback across 6 quality metrics, including Bloom's taxonomy, distractor quality, and difficulty calibration.
- Built a hybrid GraphRAG and vector retrieval layer using Neo4j, Qdrant, and LightRAG to ensure deep, traceable context grounding from raw curriculum materials.

IoT and Machine Learning Developer (Freelance Project) 2024 (Project-based)

Smart Waste Bin and Waste Management System

- Developed an IoT-based smart waste management system using computer vision; fine-tuned a MobileNetV2 model with TensorFlow and Keras achieving 88% accuracy in classifying plastic and organic waste.
- Built core Flutter mobile app features to simplify waste disposal for users and integrated real-time ML model on ESP32-CAM, including data preprocessing and system testing.

SKILLS

Programming Languages: Python, JavaScript

AI/LLM Frameworks: LangChain, LlamaIndex, Dify, Hugging Face, n8n

AI Concepts & Architectures: Retrieval-Augmented Generation, Agentic Workflows, Multi-Agent Systems, Prompt Engineering, Natural Language Processing (NLP), Fine-Tuning

Vector & Graph Databases: Qdrant, Pinecone, Neo4j, LightRAG

Backend & Web Integration: FastAPI, React.js, REST APIs

Tools & Infrastructure: Git, GitHub, Docker, Postman

PROJECTS

Enterprise Knowledge Assistant

- Built an AI Q&A agent using FastAPI and LangChain to answer document questions via a Qdrant vector database.
- Automated workflow tasks by enabling the agent to search docs, extract structured JSON data, and trigger external webhooks.
- Containerized the system with Docker and Docker Compose for simple local setup and cloud deployment.

EDUCATION & CERTIFICATIONS

BSc in Computing (4th Year Undergraduate) 2027

National Institute of Business Management, Colombo 07